Singapore School Kinderland

Secondary 1 (Sec 1) Computer Science — Term 2 Comprehensive Practice Worksheet — TEACHER ANSWER KEY

Topics: Error Detection Methods (TB Pg 54–62) & Encryption (TB Pg 63–74) | Total Marks: 40 | Detailed Marking Scheme & Solutions

Student Name: _____ **Class:** Sec 1 | **Date:** _____ **Marks:** / 40

SECTION A: PARITY CHECKS & PARITY BLOCKS (10 MARKS)

Q1. Parity Bit Calculation *[4 Marks]*

A computer system transmits 7-bit ASCII data. Add the required Parity Bit (most significant bit) for each byte below:

Byte	7-bit Data Payload	Parity System	Required Parity Bit (0 or 1)	Complete 8-bit Transmitted Byte
A	1 1 0 0 1 0 1	Even Parity	**0 (count=4)**	**0 1 1 0 0 1 0 1**
B	1 0 1 1 1 1 0	Even Parity	**1 (count=5→6)**	**1 1 0 1 1 1 1 0**
C	0 1 1 0 0 0 1	Odd Parity	**0 (count=3)**	**0 0 1 1 0 0 0 1**
D	1 1 1 0 1 1 1	Odd Parity	**1 (count=6→7)**	**1 1 1 1 0 1 1 1**

Q2. Parity Block Matrix Analysis & Auto-Correction *[6 Marks]*

A block of four bytes was transmitted using **Even Parity**. An extra parity byte was appended to the bottom:

Row	Bit 7 (Parity)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Row Status (OK / Error)
Byte 1	1	0	1	1	0	0	1	0	**OK (4 ones)**
Byte 2	0	1	1	0	1	0	1	1	**ERROR (5 ones)**
Byte 3	1	1	0	1	1	0	0	1	**OK (4 ones)**
Byte 4	1	0	0	1	0	1	1	0	**OK (4 ones)**
Parity Byte	**1**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**OK**

(a) Identify the corrupted row and column: **Byte 2** | Corrupted Column: **Bit 4** (Column 4 has 3 ones → odd) *[2]*

(b) Original bit: **0** | Corrected bit: **1** *[2]*

(c) Explain why this parity block cannot correct an error if two bits flip in the same row and column: *[2]*

Model Answer: If 2 bits flip in the same row and same column, both row parity and column parity will still register as even numbers, meaning the parity checks will report NO error and the corrupted bits cannot be identified.